Mapping of Land Cover Over Highly Heterogeneous Areas in Yunnan Province With Active and Passive Remotely Sensed Data

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***Abstract*— Yunnan Province is one of the global biodiversity conservation hotspots. Due to the fragmented terrain and spatial heterogeneity of the plateau, land cover (LC) mapping in this region has always been challenging. In this study, we integrated 1593 scenes of Sentinel-1 and 8320 scenes of Sentinel-2 (S2) imagery, proposed a sample data generation strategy based on domain information rules, and developed a two-step feature optimization method. We also constructed a tile-based classification model to mitigate the impact of spatial heterogeneity in plateau regions. By comparing the performance of 45 different combinations of predictive variables, we analyzed the effectiveness of high-dimensional predictive variable optimization. We evaluated the impact of predictive variable sets, sensor types, classifier algorithms, and sample thresholds on LC mapping. The results indicate that: 1) the two-step feature optimization method can improve the classification accuracy of machine learning (ML). With optimal features, the model's overall accuracy (OA) reached 92.21%, an increase of 0.64%–8.16% compared to using raw features alone; 2) the near-infrared band is a highly effective spectral predictor and slope significantly impacts the spatial distribution of LC in Yunnan Province; and 3) changes in sample thresholds affect the stability of prediction models and fivefold cross validation shows that under the optimal sample threshold scenario, prediction accuracy improves by 0.95%–2.76%; and 4) the tile-based classification model reduces the errors compared to traditional global models, mitigating heterogeneity effects. This study provides a new approach for classification in heterogeneous plateau regions and offers valuable data support for biodiversity conservation efforts.**

***Index Terms*— Google Earth Engine, land cover (LC) classification, tile-based classification, time series, training sample generation.**

Received 1 April 2024; revised 16 July 2024 and 26 August 2024; accepted 19 September 2024. Date of publication 26 September 2024; date of current version 4 October 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42161067 and in part by the Platform Construction Project of High Level Talent in Kunming University of Science and Technology (KUST). *(Corresponding authors: Bo-Hui Tang; Zhifang Zhao.)*

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Digital Object Identifier 10.1109/TGRS.2024.3465590

I. Introduction

LAND cover (LC) mapping plays a crucial role in the management of natural resources, offering vital information that is pertinent to scientific research in various domains, including climate change, biodiversity conservation, ecosystem preservation, and urban planning [1], [2]. In recent years, with the gradual improvement of remote sensing data resolution in both spatial and temporal dimensions, studies on LC and its changes have deepened [3]. However, in large-scale LC mapping research, spatial heterogeneity and difficulties in sample acquisition significantly constrain the accuracy and efficiency of mapping, especially in highly heterogeneous plateau regions. Therefore, improving LC mapping methods and techniques to address these challenges is particularly important.

The emergence and development of remote sensing technology have provided crucial support for large-scale LC mapping. Since 1990, numerous scholars and research institutions have developed LC products at various resolutions, including climate change initiative LC V2 (CCI-LC) [4], global LC characterization (GLCC) [5], and global LC - SHARE (GLC-SHARE) [6]. However, all of these products have low resolutions, and they come with significant limitations when used as foundational data for other studies. Since 2015, high-resolution LC products have been introduced [7]. Nevertheless, these products are influenced by factors such as data sources, classification methods, and classification schemes, and in some regions (e.g., plateaus and tropical areas), their applicability and accuracy cannot be guaranteed. Many of these products have an accuracy below 70%. Furthermore, they are only available for specific years and lack regular updates. Due to spatial heterogeneity and constraints related to the availability of remote sensing data, accurate high-resolution LC mapping in plateau regions (above 1000 m in elevation) still faces many challenges [8].

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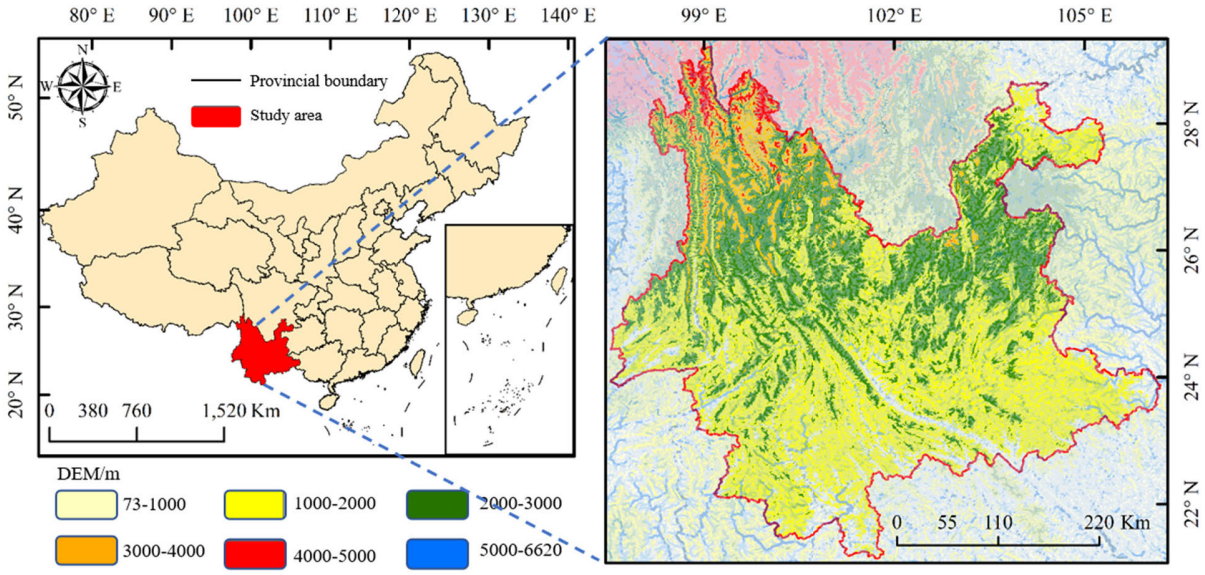


Fig. 1. Overview of the study area.

The emergence of the GEE cloud platform has enabled users to extract regional or national LC information from personal computers. It also overcomes the challenge of low optical data availability in highland areas through image synthesis. GEE has been successfully applied in various studies, including specific crop mapping, farmland mapping, and LC mapping [9], [10]. However, these supervised classification methods heavily rely on ground truth sample data, which is a major drawback in large-scale LC mapping [11]. Collecting ground truth sample data is a labor-intensive process. Although unsupervised classification does not require sample data, its application is limited in large-scale LC mapping due to its relatively poor performance [12]. Recent studies have suggested using existing high-precision LC products as a source of sample data, as they can potentially provide a reliable pool of samples. For example, Wen et al. [13] generated maize sample data based on the crop data layer product using an adaptive strategy to derive a spatial distribution map of maize in a local area of the USA. Zhang et al. [14] combined MODIS enhanced vegetation index (EVI) and VIIRS NTL imagery with the GlobeLand30 LC product to automatically generate random impervious surface sample data on a global scale. Xuan et al. [16] randomly generated sample points based on the classification results published by You et al. [15], which were used to create LC maps for rice, corn, and soybean in the Northeastern region of China. In addition, some studies have demonstrated that combining active and passive remote sensing data can provide complementary information, thereby improving the accuracy and stability of remote sensing mapping [17], [18]. However, the integration of multiple sources significantly increases the data volume, feature dimensions, computational cost, and model complexity. At the same time, due to the scale effect, there are still many challenges in conducting large-scale mapping studies in highly heterogeneous areas.

In this study, we first proposed a new method for automatically generating sample data based on domain information rules. This method significantly improves the efficiency and quality of samples for large-scale LC mapping. Subsequently, we developed a two-step feature optimization method and compared the classification performance of predictive features. Finally, we constructed a tile-based classification model to address the severe heterogeneity impacts in plateau regions. Overall, this study aims to answer the following questions.

1) How does the combination of Sentinel-1, Sentinel-2 (S2), and topographic data affect large-scale LC mapping in plateau regions?
2) Does the feature optimization method improve the accuracy of LC mapping?
3) Do the tile-based classification model, different machine learning (ML) algorithms, and the quantity of samples significantly affect the prediction results?

II. Preparation Research Area and Data

A. Research Area

The study area is situated in the southwestern region of China, spanning from latitude 21°8′ to 29°15′N and longitude 97°31′ to 106°11′E (Fig. 1). It encompasses a total area of 394 100 km^2 and features a typical mountainous plateau terrain, with mountains covering 88.64% of the total provincial area [19]. Yunnan Province boasts the highest diversity of plant and animal species in China and is renowned as the "Kingdom of Plants and Animals" and the "Kingdom of Nonferrous Metals." This region holds significant importance for ecological environmental preservation and the conservation of biodiversity. More than 90% of the province's land area is situated within the plateau region, characterized by a climate that includes cold, temperate, and hot zones, including subtropical areas.

B. Datasets and Preprocessing

1) Sentinel Data: The S2 data consist of two high-resolution satellites, 2A and 2B, with a temporal resolution

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TABLE I
LC DATA PRODUCTS USED IN THIS STUDY

Dataset	Spatial resolution	Mapping Time	Mapping Range	Total accuracy
FROM	10m	2017	global	72.76%
CLCD2015	30m	2015	china	80%
CLCD2020	30m	2020	china	80%
ESA10	10m	2020	global	74.40%
ESRI10	10m	2020	global	85.96%

of five days and a spatial resolution of 10 m, and include 13 bands. Depending on the atmospheric correction status, the data are classified into two levels: L1C and L2A. L2A data have undergone radiometric calibration and atmospheric correction. In this study, we collected 8320 scenes of L2A data captured from January 2020 to December 2020 for the classification of the base map. Considering the influence of clouds and haze, data with cloud cover of less than 10% were filtered, and a cloud mask was applied using the QA60 band. Finally, to reduce the impact of rain clouds, we normalized all S2 data using median synthesis [20].

Sentinel-1 data consist of two polar-orbiting satellites, A and B, equipped with C-band synthetic aperture radar (SAR) sensors, with a temporal resolution of six days and a spatial resolution of 10 m. Sentinel-1 data are unaffected by cloud cover and can observe continuously throughout the day and under all weather conditions. In this study, to synchronize with optical data temporally, we utilized 1593 scenes of Sentinel-1 IW GRD (S1) product data captured from January 2020 to December 2020, with polarization modes of VV and VH.

2) Available LC Products: We selected five products from four publicly released LC datasets in recent years as the sources for our baseline sample pool (Table I). The FROM_GLC10 (FROM) product is the first global 10-m LC map for 2017, created by Gong et al. [21] from Tsinghua University based on S2 temporal data, with an overall accuracy (OA) of 72.76% [22]. The China LC dataset (CLCD) covering the period from 1985 to 2020 was produced by Yang and Huang [23] from Wuhan University using Landsat data, with an OA of 80%. The ESA WorldCover 10-m (ESA10) product is a global 10-m LC dataset created jointly by European Space Agency and multiple institutions, utilizing S1 and S2 data, with an accuracy of 74.40% [7]. The ESRI10 product, with a resolution of 10 m, is sourced from the Environmental Systems Research Institute (ESRI) and generated using two data, with an OA of 85.96% [24], [25]. Although these LC datasets have different resolutions, this study only requires the spatial location information of each cover type from the data. The errors caused by different resolutions have little or no impact on this study.

3) Other Data: To assess the accuracy of the LC map, we utilized the official area parameters of various LC categories from the 2020 China's Third National Land Survey data (https://gtdc.mnr.gov.cn/) to validate our produced 10-m LC mapping results. In addition, we employed a third-party independent crowdsourced testing dataset, Geo-Wiki (http://geo-wiki.org/). Based on quality flags, we filtered all independent validation points within the study area, totaling 633 points, encompassing ten LC types.

ASTER GDEM2 elevation data are a digital elevation model (DEM) dataset produced by NASA's Terra satellite, which is a next-generation Earth-observing satellite [26]. This dataset has a resolution of 30 m and is currently widely used in research related to terrain, topography, soils, geological hazards, and so on.

III. METHODS

This study proposes an integrated method for automatically and rapidly obtaining LC maps of the subtropical and tropical regions of Yunnan Province, China, based on their typical climatic characteristics. The main workflow of this method includes: 1) data acquisition and preprocessing; 2) automatic generation of sample data; 3) feature optimization; 4) mapping of LC under 45 classification scenarios; and 5) qualitative analysis and quantitative evaluation of accuracy. As depicted in Fig. 2, steps 1, 2, 4, and 5 were completed using GEE, while step 3 was accomplished using Python 3.7.

A. Sample Pool Construction

Considering the LC distribution in Yunnan Province [27], we have defined a classification system, including forest (comprising evergreen, deciduous, and mixed), water (including lakes, rivers, and reservoirs), cropland (including paddy fields and dryland), impervious (all artificial built-up areas), grassland (natural and artificial grasslands), shrubland, and others (including wetlands, ice and snow, bare rock, and barren land). Simultaneously, we have reclassified the five LC datasets used in this study according to this system to facilitate subsequent sample generation and evaluation analysis.

1) Extraction of Stable Region 1 From Homologous Data: Based on CLCD products, we conducted overlay analysis using data from 2015 to 2020, spanning a period of five years. If the LC remains consistent in both datasets for a given area, we define that overlapping portion as "Stable area 1." The formula for stable region 1 is

$$\rho^{S1} = \rho^{CLCD2015} \cap \rho^{CLCD2020} \tag{1}$$

where S1 represents stable region 1, CLCD2015 represents 2015 data for CLCD products, and CLCD2020 is defined earlier.

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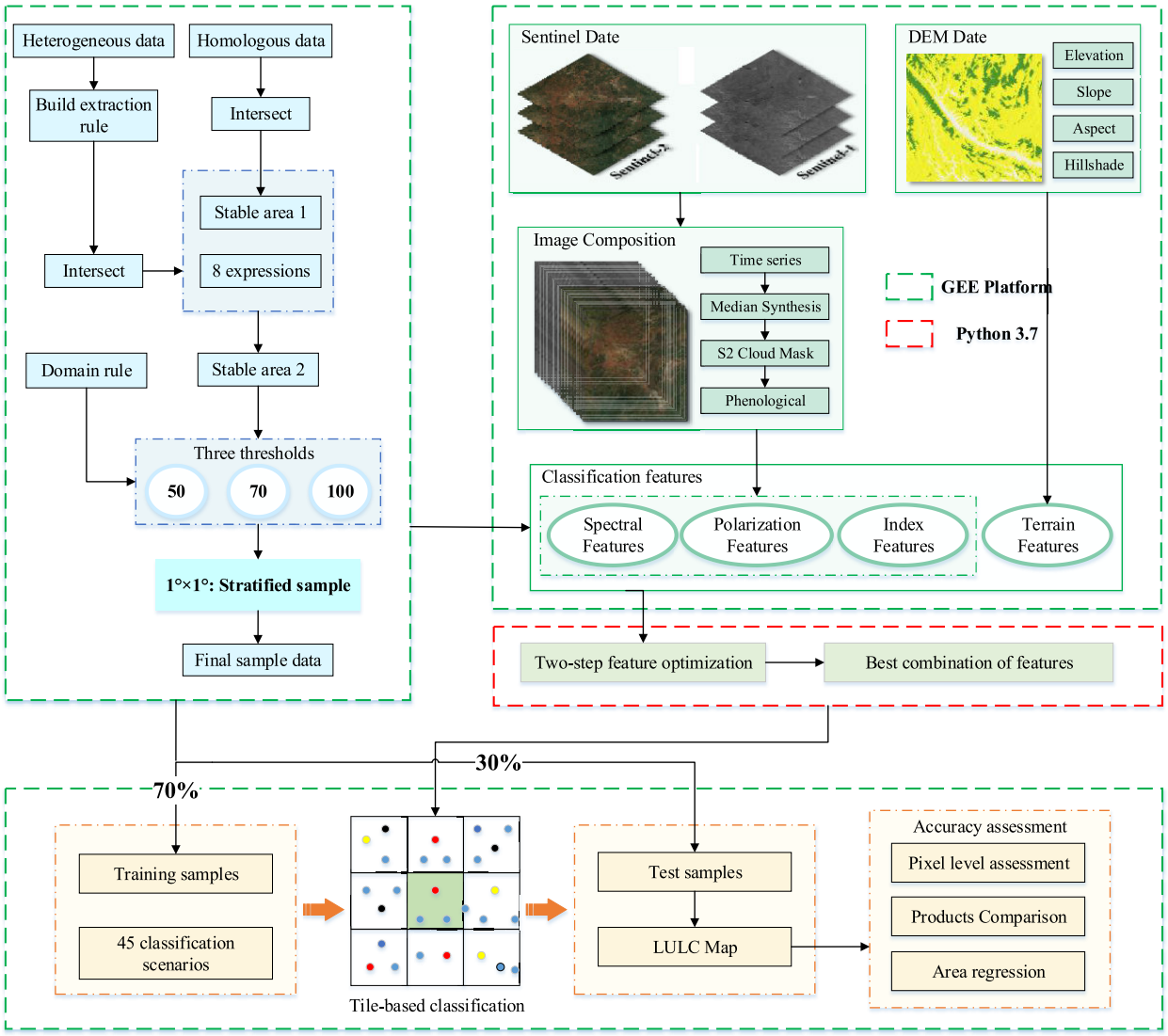


Fig. 2. Proposed flowchart.

2) Introduction of Heterogenous Data to Extract Stable Region 2: Since homogeneous data are used in Stable Region 1, it is prone to error propagation. Therefore, we introduced three heterogeneous products: FROM2017 from 2017, and ESA10 and ESRI10 from 2020. Due to the varying classification accuracy of different LC types in these products, a single mathematical expression is unlikely to be universally applicable. We developed different formulas based on the classification accuracy of these products for different LC types, differences in years, and data sources to express Stable Region 2 for different LC types. Specifically

$$\rho^{\text{Grassland}} = \rho^{\text{S1_gl}} \cap \left(\rho^{\text{ESA10_gl}} \cup \rho^{\text{ESRI10_gl}}\right) \cap \rho^{\text{FROM_gl}} \quad (2)$$

$$\rho^{\text{Shrubland}} = \left(\rho^{\text{S1_sl}} \cup \rho^{\text{ESA10_sl}}\right) \cap \rho^{\text{ESRI10_sl}} \cap \rho^{\text{FROM_sl}} \quad (3)$$

where S1_gl represents the grassland part of the stabilization region S1, S1_sl represents the shrubland part of the stabilization region S1, $\rho^{\text{Grassland}}$ represents the final stabilization region of the grassland, and the rest of the functions are the same. By comparing LC products with optical imagery, it was observed that the classification performance of the "Others" category in ESRI10 and the "Impervious" category in CLCD2020 was unsatisfactory. Consequently, these two datasets were excluded when obtaining the final training samples

$$\rho^{\text{Others}} = \rho^{\text{S1_ot}} \cap \rho^{\text{ESA10_ot}} \cap \rho^{\text{FROM_ot}} \quad (4)$$

$$\rho^{\text{IMP}} = \rho^{\text{CLCD2015_imp}} \cap \rho^{\text{ESA10_imp}} \cap \rho^{\text{ESRI10_imp}} \cap \rho^{\text{FROM_imp}} \quad (5)$$

where S1_ot represents the other parts in the stable region S1, CLCD2015_imp represents the impervious part in the CLCD2015 product, and the rest of the functions are the same. Comparison and analysis of various products with real images revealed that the classification effect of other land classes in ESRI10 product and impervious surface in CLCD2020 was not satisfactory, so these two data were excluded when obtaining the final training samples

$$\rho^{\text{Water}} = \rho^{\text{S1_wt}} \cap \rho^{\text{ESA10_wt}} \cap \rho^{\text{ESRI10_wt}} \cap \rho^{\text{FROM_wt}} \quad (6)$$

$$\rho^{\text{Forest}} = \rho^{\text{S1_fr}} \cap \rho^{\text{ESA10_fr}} \cap \rho^{\text{ESRI10_fr}} \cap \rho^{\text{FROM_fr}} \quad (7)$$

$$\rho^{\text{Cropland}} = \rho^{\text{S1_cl}} \cap \rho^{\text{ESA10_cl}} \cap \rho^{\text{ESRI10_cl}} \cap \rho^{\text{FROM_cl}} \quad (8)$$

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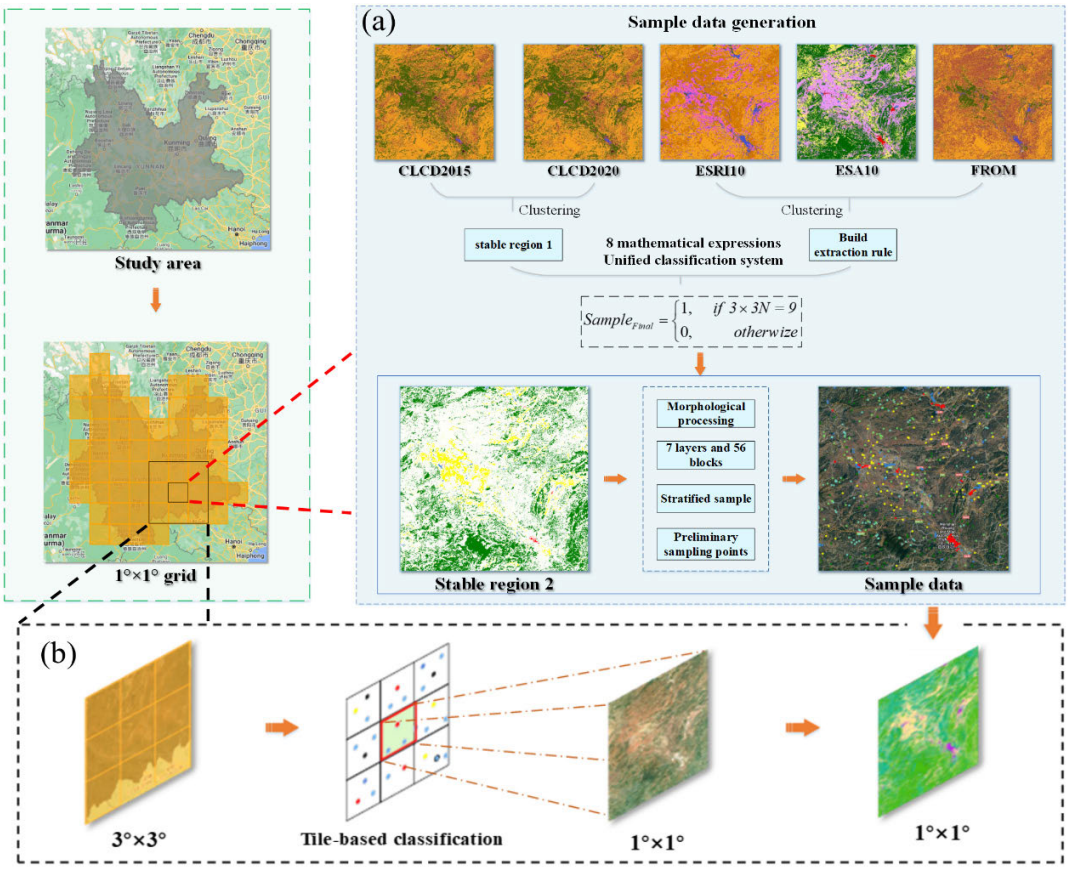


Fig. 3. (a) Stratified sampling strategy and (b) tile classification model based on spatial information weighting, with one tile representing a 1° × 1° range.

where S1_wt represents the water part of the stabilization region S1, S1_fr represents the woodland part of the stabilization region S1, S1_cl represents the cropland part of the stabilization region S1, and the rest of the functions are the same.

3) Sample Pool Generation Strategy With Spatial Information Weighting: We utilized five datasets from four products to extract stable areas. Some studies employ similar stable areas for generating sample data but overlook the randomness and discreteness of sample generation [28]. This leads to the possibility of each sample becoming an independent noise point in each classification. Therefore, we incorporated domain-specific rules into the automatic sampling process, stipulating that within the 3 × 3 neighborhood (3 × 3N) of each automatically selected sample data, all nine pixels must belong to the same category [Fig. 3(a)]. The rules are as follows:

$$\text{Sample}_{\text{Final}} = \begin{cases} 1, & \text{if } 3 \times 3\text{N} = 9 \\ 0, & \text{otherwize} \end{cases} \tag{9}$$

where $\text{Sample}_{\text{Final}}$ represents the generated final sample. This method can also resolve issues of exceeding operational limits in GEE due to the vast size of the study area and the large quantity of satellite data.

B. Image Feature Construction and Optimization

1) Construction of Candidate Features: Combining the ecological environment and phenological conditions of the study area, we constructed five feature spaces, encompassing a total of 40 variables (Table II). The normalized difference vegetation index (NDVI), EVI, ratio vegetation index (RVI), and difference vegetation index (DVI) capture the greenness variations during vegetation growth [29]. The normalized difference water index (NDWI), land surface water index (LSWI), soil adjusted vegetation index (SAVI), and modified NDWI (MNDWI) are beneficial for distinguishing irrigated croplands and water bodies [30]. The normalized difference built-up index (NDBI) is used to differentiate impervious surfaces [31]. Numerous studies have shown that extracting the VV and VH backscattering coefficients from radar data can improve the classification accuracy [10]. Considering the complex planting structure and fragmented farmland in the study area, adding texture features can better describe the randomness and detail of textures. We calculated four texture heterogeneity features based on the gray level co-occurrence matrix (GLCM): contrast (Con), angular second moment (Asm), correlation (Corr), and entropy (Ent). In addition, given that the study area is located in a plateau region with significant elevation variations, we introduced topographic features to explore their classification potential.

2) Two-Step Feature Optimization: In the classification process, having too many features may lead to redundancy, while having too few may result in inadequate representativeness. The introduction of each feature may cause differences in classification results and cumulative errors. Our feature optimization consists of two steps: 1) removing highly correlated features and 2) implementing the recursive feature elimination

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TABLE II
FEATURE SPACE CATEGORY AND RELATED VARIABLE CALCULATION FORMULA

Sensor Used	Feature Space	Feature Variables	Total
S2	Spectral	B1, B2, B3, B4, B5, B6, B7, B8, B8A, B9, B11, B12	12
	Index	$NDVI = (B8 - B4)/(B8 + B4)$	10
		$NDWI = (B3 - B8)/(B3 + B8)$	
		$NDWI1 = (B3 - B8A)/(B3 + B8A)$	
		$EVI = 2.5(B8 - B4)/(B8 + 6B4 - 7.5B2 + 1)$	
		$LSWI = (B8 - B12)/(B8 + B12)$	
		$NDBI = (B12 - B8)/(B12 + B8)$	
		$RVI = B8A/B4$	
		$DVI = B8A - B4$	
		$SAVI = (B8 - B4)(1 + 0.5)/(B8 + B4 + 0.5)$	
		$MNDWI = (B3 - B11)/(B3 + B11)$	
	Texture	Extraction of texture features in the B8	4
S1	Polarization	VV, VH	2
	Texture	Extraction of texture features in the VV	4
		Extraction of texture features in the VH	4
DEM	Terrain	Slope, Elevation, Hillshade, Aspect	4

(RFE) algorithm using random forest (RF). In the first step, we calculated the correlation of all features listed in Table II. Features with high correlation ($r \geq 0.90$) were removed. In the second step, we utilized the RFE algorithm to identify the most influential predictor variables [32]. Initially, we fit an ensemble RF model using all remaining predictor variables from the first step. Each predictor variable was ranked based on its importance value in the RF model. Then, the algorithm iteratively removed one predictor variable with lower importance until only a single predictor variable remained as input. During each iteration, the importance of each predictor variable in the RF model was recalculated using the retained feature set. These two steps address the interdependencies among features and retain the most important ones. Recognizing that model performance can exhibit uncertain noise when involving numerous variables, we used fivefold cross validation to evaluate the model performance. We systematically compared the prediction accuracy of each iteration and determined the final predictor variables for each prediction scenario (Table III).

C. Image Classification

1) Tile-Based Classification Model: The study area is located in a plateau region with a maximum elevation difference of 6547 m. The northern part experiences a subtropical monsoon climate, while the southern part has a tropical monsoon climate. Attempting to use sample data from the northern region to extract LC information for the southern region may lead to significant errors. Considering the differences in neighboring land features and the wide climatic range within the study area, we designed a tile-based quadrilateral classification model. The specific principles are as follows: 1) based on the latitude and longitude information of the study area, we extracted a grid of 1° × 1°, resulting in a total of 56 grids; 2) for each grid, we extracted sample information of all LC types within a 3° × 3° range centered on the grid; 3) we used the sample information from the central domain and its surrounding eight neighboring domains to classify the central grid, repeating this process for each grid, totaling 56×; and 4) based on the 56 classification results, we used the mosaic() function to stitch the results together to obtain the LC mapping of the study area [Fig. 3(b)].

2) Construction of Prediction Scenarios: To investigate the influence of sample data quantity on classification results, we set three thresholds: 50, 70, and 100 (i.e., the number of samples generated per class per grid). As shown in Table IV, there is a certain deviation between the actual and theoretical sample quantities. This is because the distribution of each LC class within each 1° × 1° grid is not uniform, leading to differences in sample generation quantities. In addition, manual corrections were made to the generated sample data, resulting in the removal of some samples. Furthermore, some sample data were lost due to system errors when loading the generated sample data into the GEE personal assets.

To investigate the impact of different sample sizes, ML methods, sensors, and feature numbers on LC mapping, a total of 45 classification scenarios were designed in this study (Table III). Each scenario was categorized based on different predictor variables, including both original and optimized

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TABLE III
DESCRIPTION OF VARIABLES INCLUDED IN EACH PREDICTION SCENARIO UNDER THREE SAMPLE THRESHOLD CONDITIONS

Sample thresholds	Scenario names	Feature Description	Number of features	
			Total	Optimized
50	R1, C1, S1	S1 Polarization	2	2
	R2, C2, S2	S1 (Polarization + Texture)	10	6
	R3, C3. S3	S2 (Spectral + Index)	22	20
	R4, C4, S4	S2 (Spectral + Index + Texture)	26	20
	R5, C5, S5	All multi-source features	40	21
70	R6, C6, S6	S1 Polarization	2	2
	R7, C7, S7	S1 (Polarization + Texture)	10	9
	R8, C8, S8	S2 (Spectral + Index)	22	20
	R9, C9, S9	S2 (Spectral + Index + Texture)	26	22
	R10, C1, S10	All multi-source features	40	20
100	R11, C11, S11	S1 Polarization	2	2
	R12, C12, S12	S1 (Polarization + Texture)	10	9
	R13, C13, S13	S2 (Spectral + Index)	22	20
	R14, C14, S14	S2 (Spectral + Index + Texture)	26	21
	R15, C15, S15	All multi-source features	40	22

R1, C1, S1: classification scenario numbers using RF, CART, and SVM ML models, respectively; other numbers follow the same logic; Total: the total number of predictive features before optimization; Optimized: The number of final predictor variables after the two-step feature optimization method.

TABLE IV
NUMBER OF SAMPLE POOLS

Threshold	Theoretical sample number	Actual sample number
50	19600	14625
70	27440	20134
100	39200	28308

Theoretical sample number: threshold × number of grids (56) × number of feature classes (7)

features. The tested classifiers included RF, support vector machine (SVM), and decision tree (CART), aiming to evaluate the optimal classifier for LC mapping. CART and SVM were used with default parameters, while the number of trees (Ntree) for RF was set to 100 [33].

D. Accuracy Evaluation

Typically, reliable statistical data are used to evaluate the accuracy of classification results. In this study, we used land survey data published by the government to assess the accuracy of the final mapping results, calculating the coefficient of determination (R^2). We then utilized the pixelArea function in GEE to calculate the coverage area for each class. In addition, we conducted a spatial consistency assessment, performing per-pixel validation and quantitative analysis based on validation samples to calculate OA, producer's accuracy (PA), user's accuracy (UA), and $F1$ score. Finally, we qualitatively compared the final classified map with existing LC products.

IV. RESULTS AND ANALYSIS

A. Model Performance

Fig. 4 shows the OA of the 45 classification scenarios. The results indicate that LC mapping results are influenced by the number and type of predictor variables, sensor type, classifier algorithm, and sample threshold. By comparing the prediction performance of raw features (solid lines) and optimized features (dashed lines), we found that using optimized features in classification always results in higher OA than using raw features. This demonstrates the positive impact of our two-step feature optimization method on LC mapping. Overall, there is a positive correlation between sample threshold and prediction accuracy; as the number of samples increases, the trained model becomes more robust. The accuracy with a sample threshold of 100 (yellow) is generally higher than that with thresholds of 70 (green) and 50 (blue), with prediction accuracy improvements ranging from 0.95% to 2.76% under the optimal sample threshold (100) scenario. Among different ML models, RF consistently achieves the highest prediction accuracy. The best prediction accuracy was found in R15 (OA: 92.21%), with OA improving by 4.35% and 7.36% points compared to C15 (OA: 87.86%) and S15 (OA: 84.85%), respectively.

Sensor type significantly impacts prediction accuracy, with S2 data showing significantly higher accuracy than S1 data, possibly explaining why S1 data are primarily used as supplementary data in classifications. As expected, combining multisource data significantly improves prediction accuracy; regardless of R15, R10, or R5, the accuracy is higher than in

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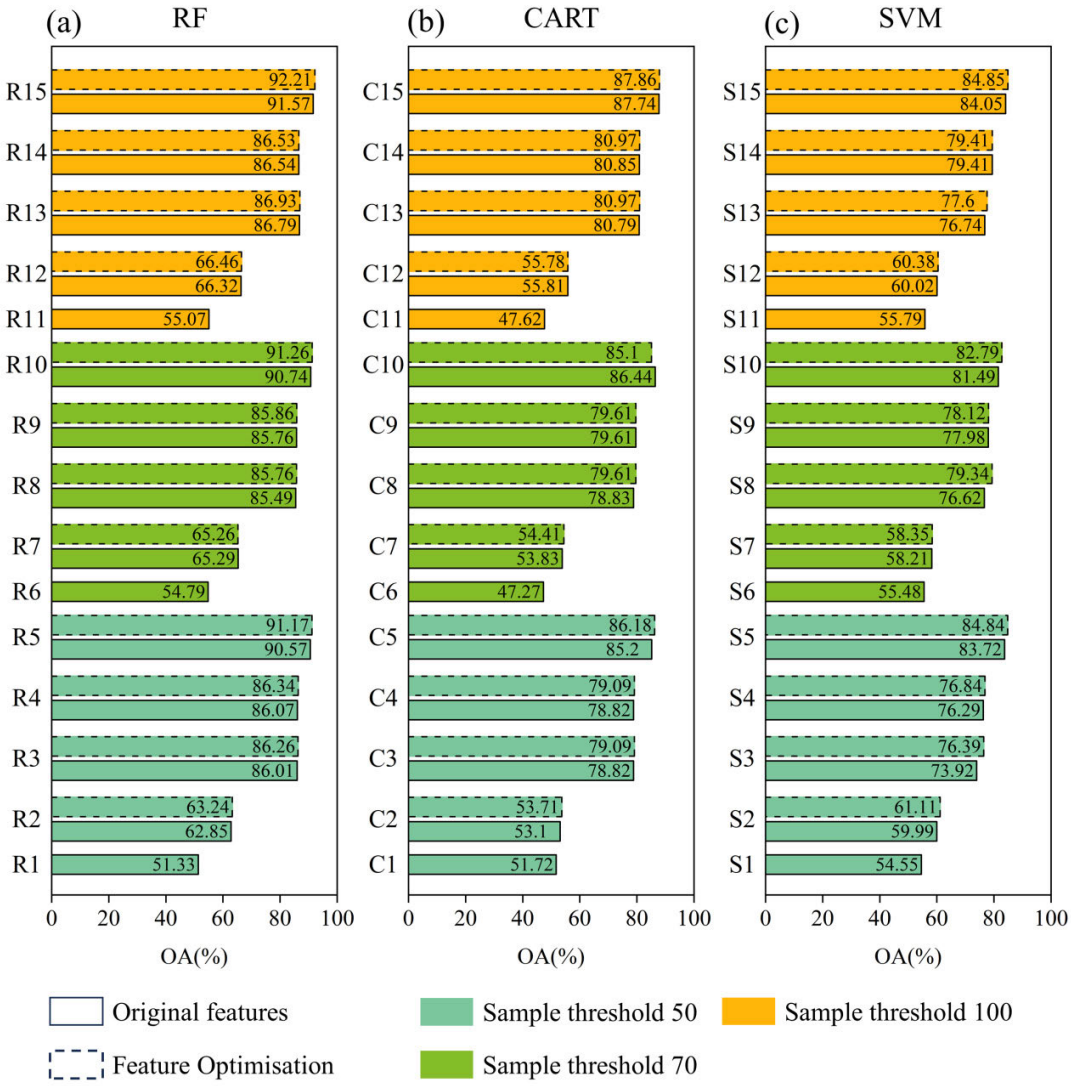


Fig. 4. Prediction accuracy under different sample thresholds and classifiers for the 45 classification scenarios. (a) RF. (b) CART. (c) SVM.

other classification scenarios with the same sample threshold. Overall, we identified the highest classification scenario (R15 with optimized features), the best classifier (RF), the optimal sample threshold (100), and the sensor contributing the most to classification (S2).

B. Feature Importance Analysis

To evaluate the performance of optimal classification features corresponding to sample thresholds of 50, 70, and 100, we used fivefold cross validation to assess R5 [Fig. 5(a) and (b)], R10 [Fig. 5(c) and (d)], and R15 [Fig. 5(e) and (f)]. The results indicate that prediction accuracy increases with the number of features, eventually stabilizing. Further increasing the number of features slightly reduces prediction accuracy, likely due to model overfitting, which lowers the test accuracy and affects the classification performance. The optimal number of features is 21 for R5, 20 for R10, and 22 for R15. The ranking of optimal feature contributions in each scenario shows that topographic features (slope) have a high weight in classification consistent with findings from other studies [34]. In addition, we observed that introducing optical texture features did not significantly improve prediction accuracy and even slightly reduced it in some scenarios (e.g., R13 and R14, and S13 and S14). This may be due to the irregular changes in pixel values caused by the fragmented terrain in the plateau region, affecting the GLCM calculation of texture features involving surrounding neighboring pixels. On the other hand, texture heterogeneity prediction factors extracted from S1's VV and VH showed a positive impact on prediction accuracy (R12 improved by 11.39% points compared to R11). However, the accuracy of using S1 data alone for mapping is concerning (R12: 66.46%). These findings underscore the importance of selecting optimal feature sets and highlight the potential pitfalls of feature redundancy and overfitting, particularly in complex terrain areas such as the Yunnan Plateau.

C. Improved LC Map of Yunnan Province

The confusion matrix for the mapping results, as shown in Table V (R15), reveals that the UA and PA for grassland,

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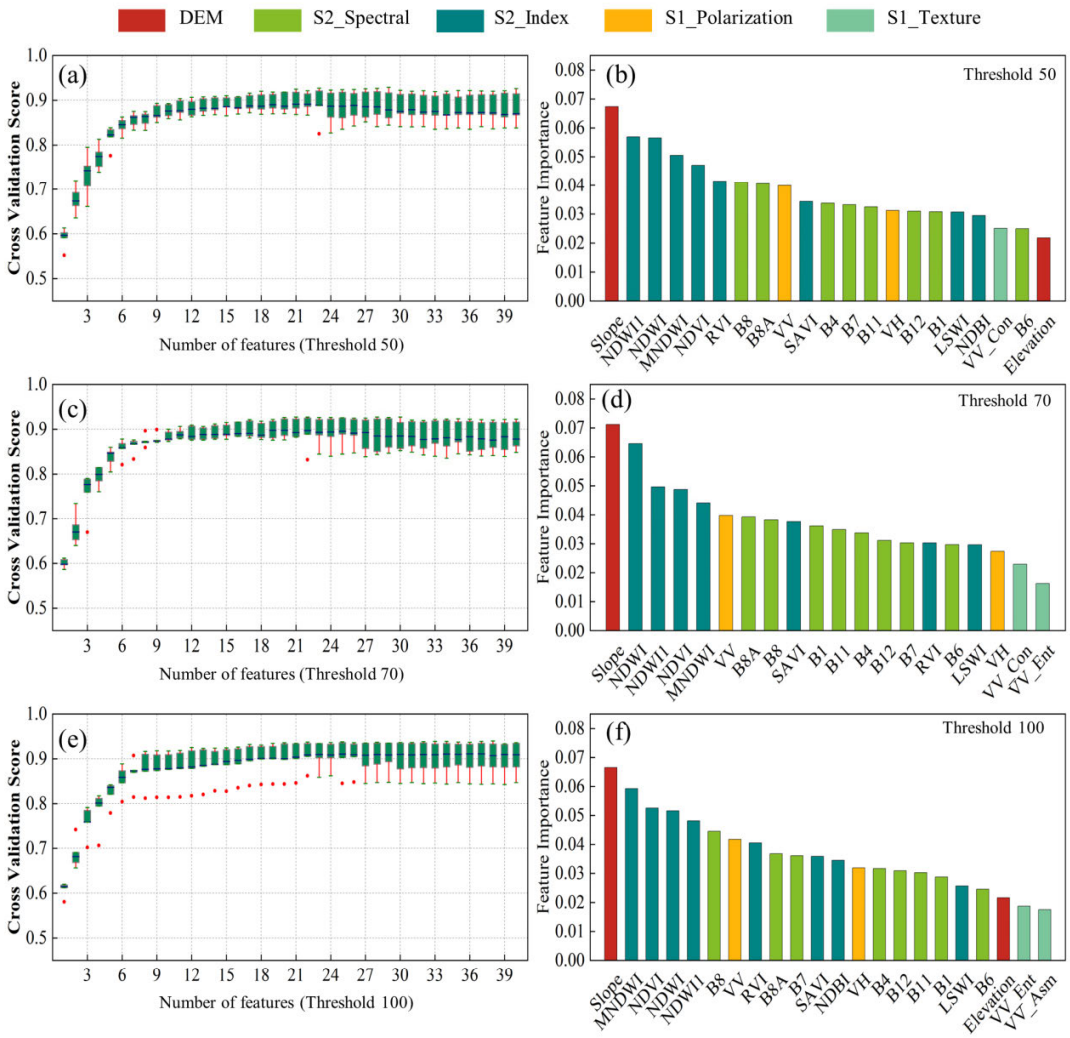


Fig. 5. Variation of test accuracy with the number of features and feature importance. (a), (c), and (e) Trend of test accuracy with an increasing number of features. (b), (d), and (f) Optimal feature importance ranking for the three sample thresholds.

forest, cropland, water, impervious, and others are all above 0.8. In particular, the PA for water is relatively high, at 98.51%. This is because water-related features have a significant contribution to classification (Fig. 5). There is noticeable confusion between shrubland and grassland, resulting in a slightly lower UA for shrubland. Overall, the OA reaches 92.21%, and the stricter $F1$ score reaches 89.30%, which represents relatively high accuracy for a 10-m provincial-level LC map. Fig. 6 illustrates the 10-m LC map of Yunnan Province in 2020, created using a combination of GEE and Python 3.7. The map demonstrates good spatial consistency across the entire province. The proportions of each LC category are as follows: grassland occupies 19.23%, forest occupies 59.16%, cropland occupies 15.45%, water occupies 1.07%, impervious occupies 1.98%, shrubland occupies 2.36%, and others occupy 0.75%.

V. Validation and Discussion

A. Accuracy Assessment of YLCD

As shown in Fig. 7, ESRI10 has a substantial amount of misclassification in recognizing impervious surfaces, leading to inconsistent detailed representation. Although it achieves an OA of 85%, the accuracy may be overestimated [Fig. 7(c)]. CLCD2020 and FROM exhibit a considerable number of omissions in recognizing water bodies [Fig. 7(f)], while ESRI10 performs better in water body recognition but still has a substantial amount of misclassification in impervious surfaces. Overall, the four products, CLCD2020, FROM, ESA10, and ESRI10, show low consistency in LC information in Yunnan Province. This may be due to the high spatial heterogeneity and complex surface in the plateau region [35], [36]. In contrast, the LC map obtained by our method provides visually satisfying descriptions.

For an objective quantitative analysis of YLCD accuracy, we utilized the third-party independent validation dataset Geowiki (2011–2012). As shown in Table VI, the $F1$ scores for all five product forested areas reached 67%, exhibiting relatively stable changes over a ten-year period compared to other land types. This consistency may be attributed to the fact that forested lands in the subtropical highlands are predominantly situated at high altitudes with undulating topography. These areas are highly resilient to disturbances,

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TABLE V
CONFUSION MATRIX OF THE 10-m LC MAP OF YUNNAN PROVINCE

Type	Grassland	Forest	Cropland	Water	Impervious	Shrubland	others	Total	UA(%)
Grassland	1374	42	102	4	7	24	7	1560	88.08
Forest	64	1565	25	2	0	15	0	1671	93.66
Cropland	124	28	1425	3	11	0	0	1591	89.57
Water	3	5	15	1590	4	0	8	1625	97.85
Impervious	2	1	10	4	1421	0	7	1445	98.34
Shrubland	46	34	1	0	0	154	0	235	65.53
others	14	8	20	11	13	0	326	392	83.16
Total	1627	1683	1598	1614	1456	193	348	8519	
PA(%)	84.45	92.99	89.17	98.51	97.60	79.79	93.68		

mUA = 88.03%, mPA = 90.88%, mF_1 = 89.30%, OA = 92.21%

mUA represents mean user accuracy, mPA represents mean production accuracy, and mF1 represents mean F1 score.

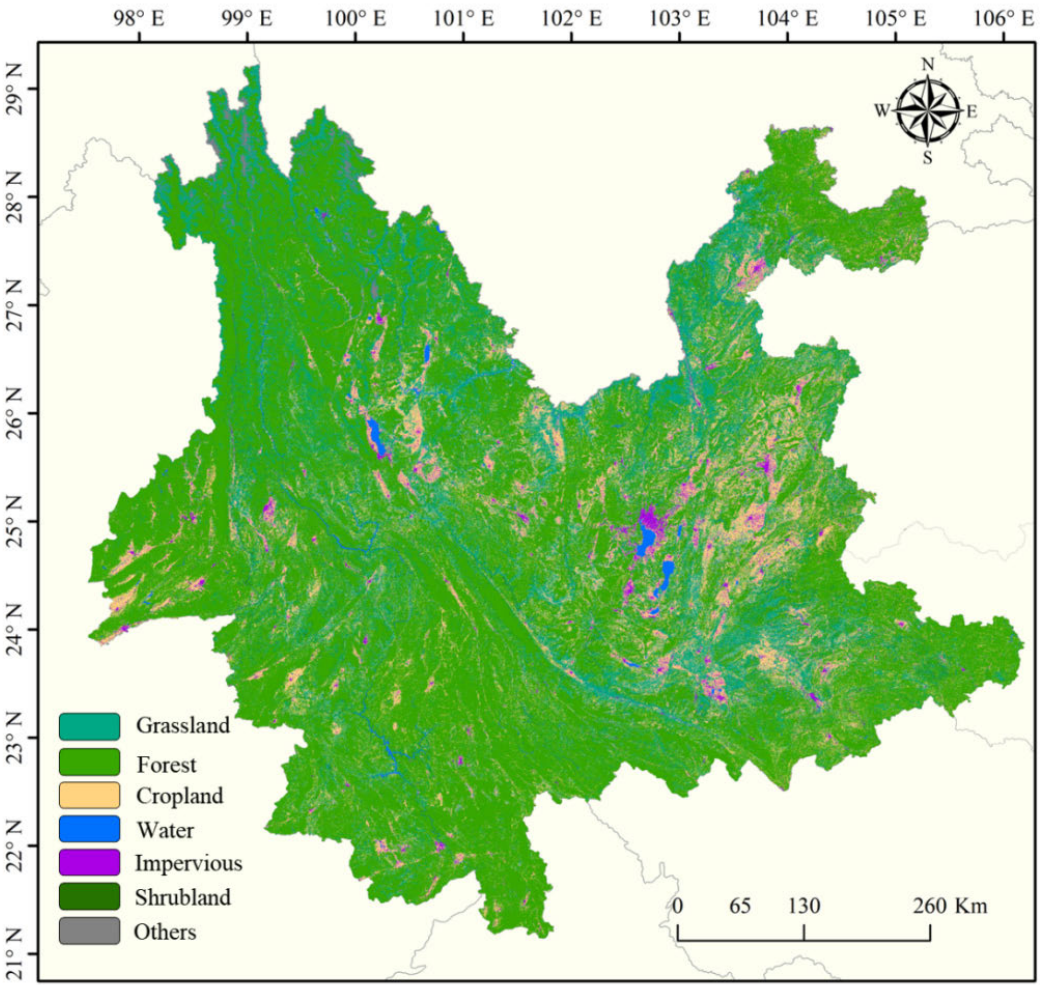


Fig. 6. Improved 2020 Yunnan LC data (YLCD).

experience low cyclical variability, and are less affected by human activities. While UA and PA were not consistently optimal for every YLCD land class, grassland, forest, and water consistently achieved the highest $F1$ scores. Overall, YLCD (50.08%) exhibited the highest OA relative to FROM (47.39%), CLCD2020 (48.34%), ESA10 (48.34%), and ESRI10 (45.50%). When compared to all historical LC products, YLCD demonstrated more accurate classification results and higher overall classification accuracy.

Fig. 8 shows the comparison of area estimates at the municipal and county levels between the 2020 Third National Land Survey data of China and our product. Since the Third National

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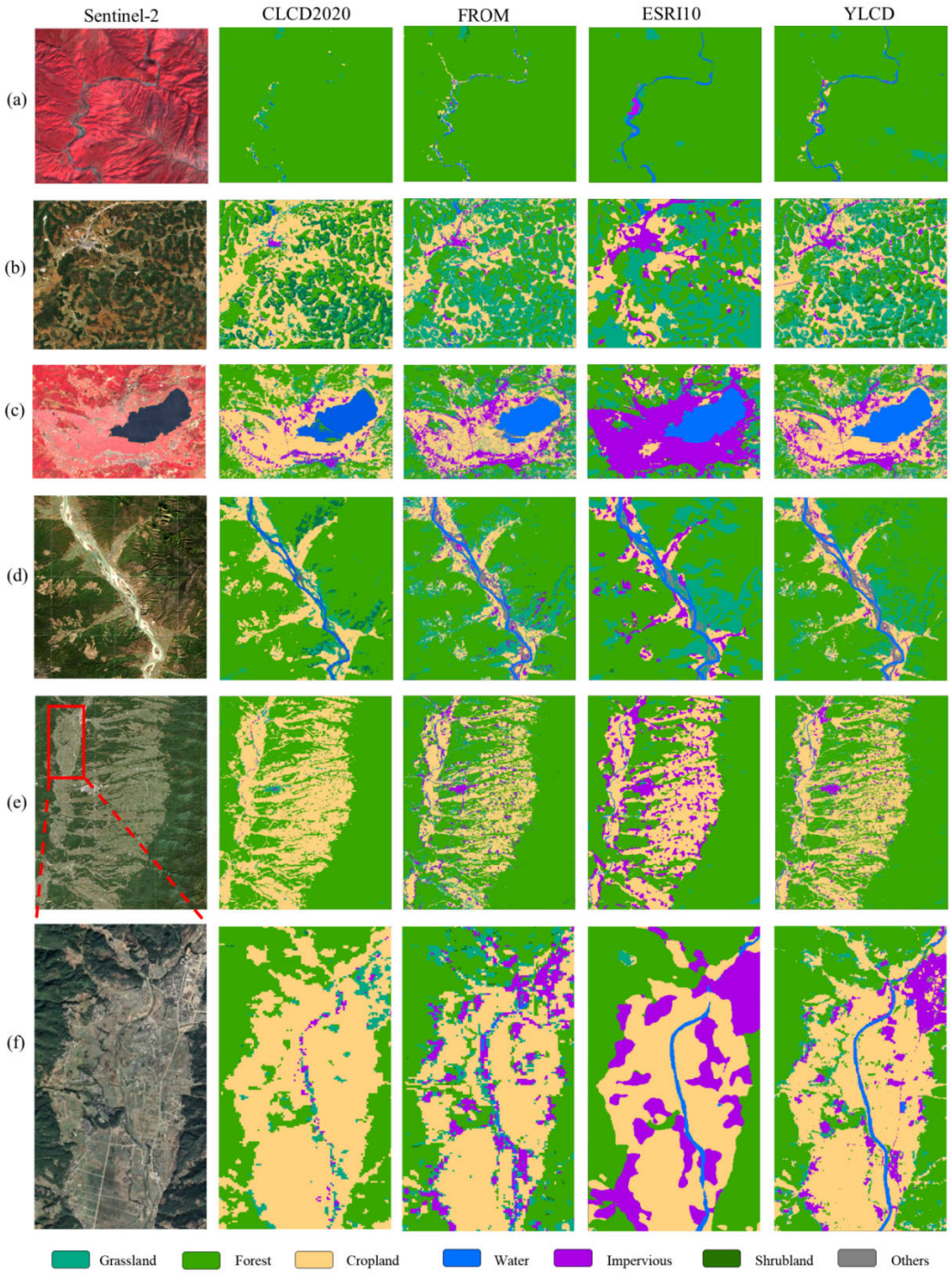


Fig. 7. Comparison of YLCD with other LC products. (a)–(f) Zoomed-in views of sub-regions.

Land Survey data did not include the shrubland category, we conducted statistical analyses for water, impervious, cropland, and forest. The results indicate that YLCD has reliable accuracy, with correlation coefficients for water, impervious, cropland, and forest all exceeding 0.9. The comparison reveals that the areas of cropland and forest are slightly underestimated compared to the survey data. This underestimation is partly due to the effects of heterogeneity and data resolution limitations, where some rain-fed plants are easily confused with shrubs. In addition, the official survey data tend to overestimate these areas due to large-scale statistical methods that often overlook growth differences within the same crop and the discrete distribution of different land features.

B. Advantages of the Proposed Method

Our results indicate that the combination of multisource data is highly suitable for large-scale LC mapping in plateau regions, consistent with many cartographic studies [37]. The highlights of our research are: 1) development of a sample

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TABLE VI
COMPARISON OF DIFFERENT PRODUCTS BASED ON GEO-WIKI'S INDEPENDENT TEST SET

LC Products		Grassland	Forest	Cropland	Water	Impervious	Shrubland	Others	OA (%)
FROM	UA	16.55	59.23	55.06	66.67	44.44	5.26	100.00	47.39
	PA	30.26	78.47	55.06	28.57	72.73	2.08	1.56	
	F1	21.40	67.50	55.06	40.00	55.17	2.99	3.08	
CLCD2020	UA	6.45	59.84	33.90	66.67	66.67	20.00	1.00	48.34
	PA	2.63	83.21	67.42	28.57	54.55	12.50	1.56	
	F1	3.74	69.62	45.11	40.00	60.00	15.38	3.08	
ESA10	UA	23.97	61.00	38.32	100.00	72.73	0.00	27.27	48.34
	PA	38.16	79.93	46.07	28.57	72.73	0.00	7.03	
	F1	29.44	69.19	41.84	44.44	72.73	0.00	11.18	
ESRI10	UA	0.00	62.22	54.39	66.67	13.92	54.39	66.67	45.50
	PA	0.00	81.75	34.83	28.57	100.00	34.83	3.12	
	F1	0.00	70.66	42.47	40.00	24.44	42.47	5.97	
YLCD	UA	28.67	66.03	36.80	100.00	40.91	30.00	66.67	50.08
	PA	56.58	76.64	51.69	28.57	81.82	6.25	3.13	
	F1	38.05	70.95	42.99	44.44	54.55	10.34	5.97	

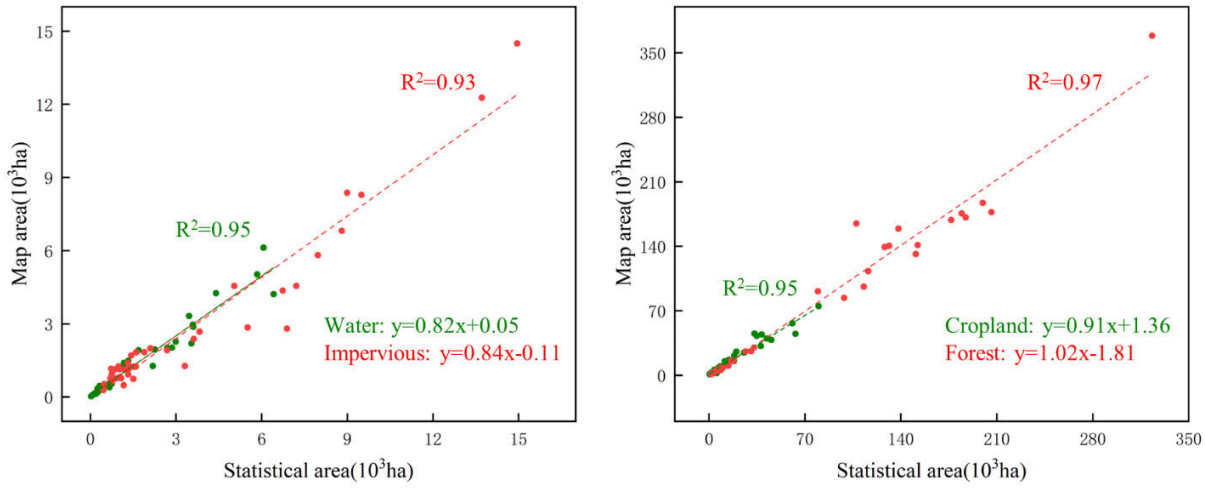


Fig. 8. Regression analysis of the third national land survey data and YCLD area.

set generation strategy that considers spatial information; 2) development of a two-step feature optimization method; and 3) construction of a tile-based classification model that incorporates domain information.

First, most croplands in Yunnan Province are concentrated in the eastern regions, typically at elevations below 2000 m (Fig. 1), due to easier transportation and cultivation at lower altitudes. In addition, areas rich in croplands are often associated with a significant amount of Impervious, which explains why the sample distribution for impervious and cropland is denser compared to other regions [Fig. 9(b) and (c)]. However, uneven distribution of training sample data in geographical space can lead to false classification accuracy [38]. Our proposed sample generation strategy creates spatial position rules that improve the issue of local clustering of sample data [Fig. 9(d)–(f)]. This strategy ensures that all 9 pixels within a 3×3 neighborhood must belong to the same category for the center pixel to be defined as a sample, ensuring spatial consistency and classification purity.

In addition, combining multisource data increases the feature dimensions. This study addresses the computational complexity and low efficiency caused by feature redundancy by incorporating a two-step feature optimization method during model training to iteratively obtain the optimal combination. During classification, the importance of samples may surpass that of features. For example, in scenario R3, the sample threshold is 50, while in scenario R13, the sample threshold is 100. Under equivalent conditions, the average accuracy for shrubland in R13 improved from 0.649 to 0.714 [Fig. 10(a) and (b)]. Even with a few erroneous training samples, RF can achieve stable classification due to its robustness [21]. The optimal scenario R15 has 22 optimized features, with OA improved by 0.64% points compared to preoptimization. The model runtime decreased from 3950 to 1997 ms, enhancing operational efficiency by 1.98×.

Finally, in large-scale LC mapping, errors among different LC types are common, especially in highland areas with fragmented plots and high heterogeneity [39]. In addition, the issue

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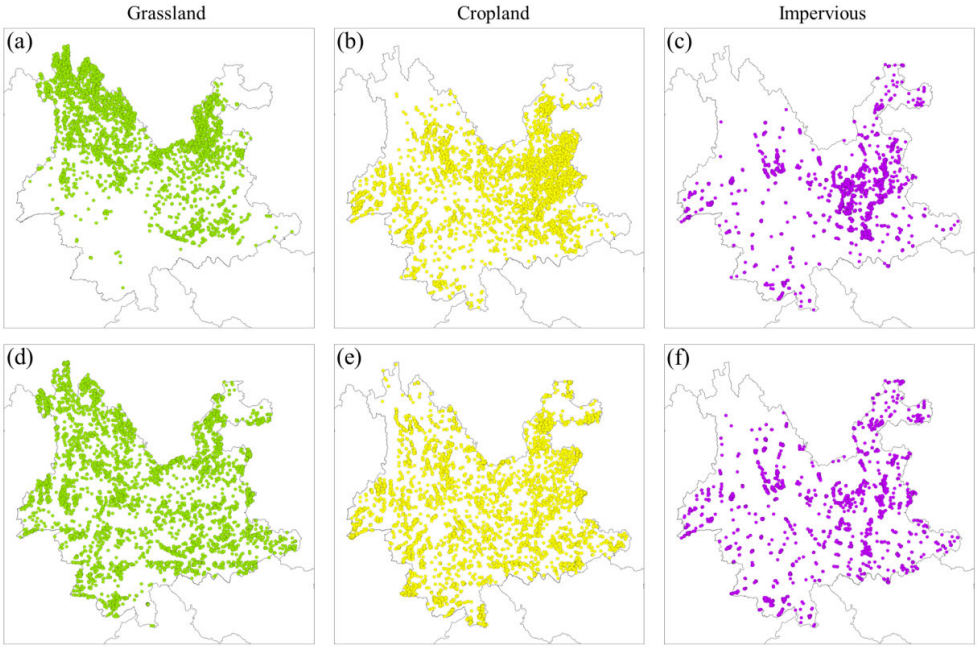


Fig. 9. Comparison of automatic generation of sample data. (a)–(c) Traditional automatic sample generation method. (d)–(f) Quadrilateral-based stratified sampling strategy.

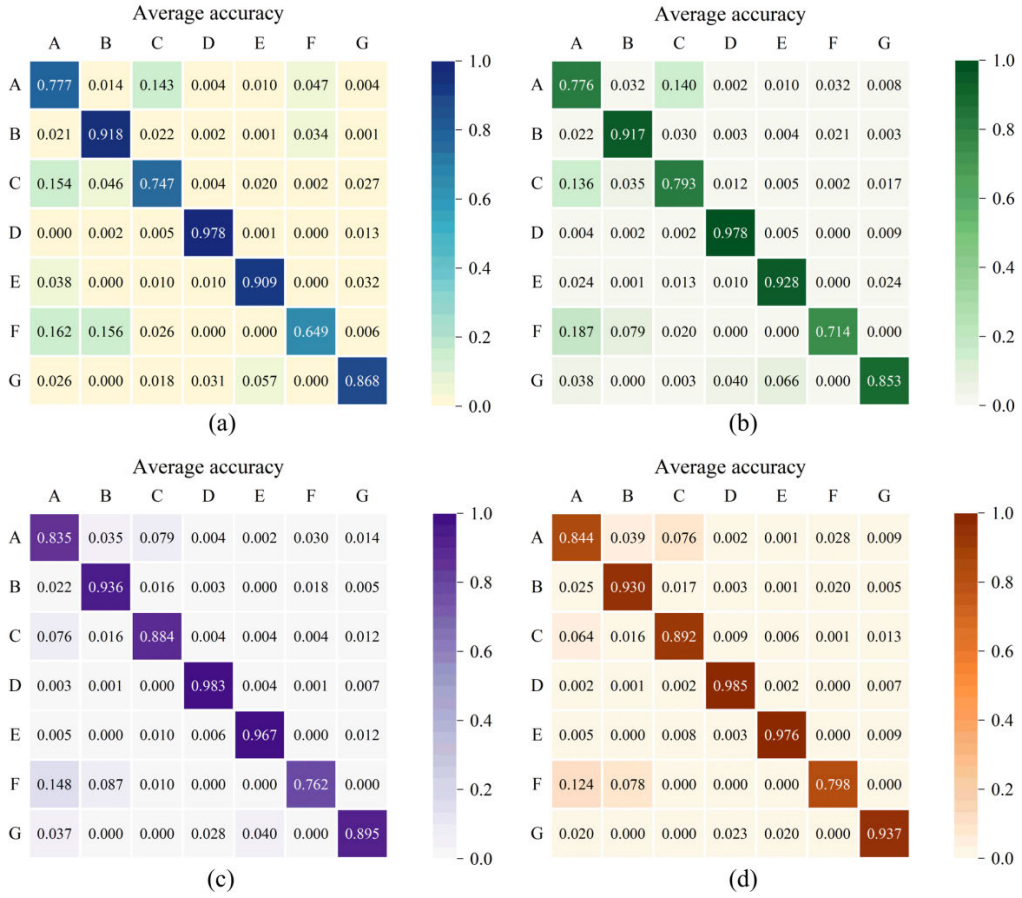


Fig. 10. Heatmap of average accuracy using different sample thresholds with feature optimization. (a) Feature optimization R3. (b) Feature optimization R13. (c) Raw features R15. (d) Feature optimization R15. A: grassland; B: forest; C: cropland; D: water; E: impervious; F: shrubland; and G: others.

of sample imbalance in classification may lead to an inflated OA [40], with categories having fewer samples (e.g., shrubland and others) performing relatively poorly. Thus, in global-based classification, there is a noticeable misclassification between

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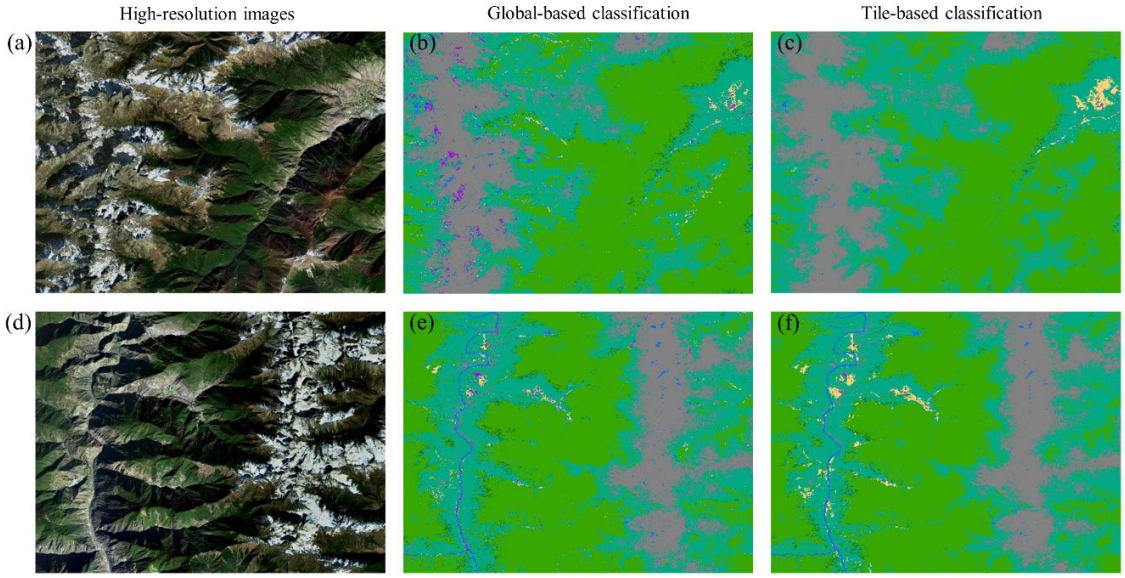


Fig. 11. Local comparison between tile-based classification and global-based classification. (a) and (d) Google Earth imagery. (b) and (e) Results of the global classification. (c) and (f) Results of the tile-based classification.

impervious and others, and between grassland and cropland in higher altitude regions [Fig. 11(b) and (e)]. In contrast, these misclassification issues are significantly improved in tile-based classification [Fig. 11(c) and (f)]. This indicates that the tile-based classification model considers the similarity and specificity between adjacent LCs, thereby improving the spatial heterogeneity issue. It also reduces the relative proportion of sample numbers in the input model, partially addressing the sample imbalance problem. The global-based classification does not possess these advantages. Ultimately, the PA for grassland and shrubland reached 84.85% and 79.79%, respectively. Although challenges remain in classifying shrubland, the accuracy has improved significantly.

C. Uncertainty

Although the final classification results are reliable, there are still some uncertainties, and the potential reasons are given as follows.

1) *Plateau Heterogeneity:* Although our method proves to improve the impact of heterogeneity, it cannot completely solve it. In some LC types where spectral features are prone to overlap (such as shrubland and grassland), there are still some errors. Therefore, future research suggests introducing higher resolution data and more heterogeneity indicators (such as more comprehensive temporal texture features) to improve the classification results of LC map change detection.
2) *LC Type Limitations:* In areas where certain LC types are limited and sparsely distributed, the final samples may not be representative. This issue can be alleviated through visual interpretation and field surveys in these sparse areas.
3) *Classifier Limitations:* The GEE platform offers various ML algorithms, but more recent algorithms, such as deep learning (DL), are relatively scarce, even though DL algorithms could potentially provide higher classification performance [41]. Although DL algorithms typically require greater computational resources, which may limit their practicality in large-scale applications, we plan to implement DL models on the GEE platform in the future. This will enable us to handle large-scale heterogeneous datasets and optimize the use of computational resources to achieve more accurate LC mapping results [42].

VI. Conclusion

In this study, we combined active and passive remote sensing data and proposed a novel method for mapping LC in the highly heterogeneous regions of Yunnan Province. This method integrates a set of automated strategies for generating sample data, employs a tile-based classification model, and incorporates heterogeneity indices to produce a 10-m LC map of Yunnan Province. The results show that the following conditions hold.

1) The two-step feature optimization method can improve the classification capability and operational efficiency of ML in multidimensional datasets, reducing data redundancy and the probability of overfitting. Under the feature optimization scenario, the OA growth rate is about 0.1%–2.72%.
2) Compared to global classification, the tile-based classification model considers the correlation of neighboring land features, which can mitigate the impact of heterogeneity and enhance LC mapping capability.
3) The near-infrared band is the most advantageous spectral band for predicting LC, and the slope significantly impacts the spatial distribution of LC in Yunnan Province.
4) Features extracted from S2 data (index features and spectral features) have the highest classification weights, while predictor factors from S1 data contribute the least.

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Notably, the inclusion of four S2 texture features in this study has a minimal impact on or even slightly decreases the final prediction accuracy, whereas S1 texture features positively impact prediction accuracy. Overall, this study provides a new approach for large-scale LC mapping in complex and highly heterogeneous plateau regions, demonstrating significant potential.

ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers and the editors for their valuable comments and suggestions that improved the manuscript. They also thank *GEE study Room* public account for the help.

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